

A MINI PROJECT REPORT ON

STUDENT GRADE ANALYZER USING R PROGRAMMING

Submitted to partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING (DATA SCIENCE)

By

G.SPANDHANA REDDY----- (24911A6783)

L.SAGAR -----(24911A6798)

M.AKSHAYA REDDY----- (24911A67A2)

M.SANJANA----- (24911A67A6)

Under the Guidance of

B.SUNITHA



Department of Computer Science & Engineering (Data Science)

VIDYA JYOTHI INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

(Approved by AICTE , Accredited by NAAC, NBA & permanently Affiliated to JNTUH,)

Aziz Nagar Gate, C.B. Post, Hyderabad-500075

2025-2026.

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(DATA SCIENCE)**

CERTIFICATE

This is to certify that the project report titled “STUDENT GRADE ANALYZER USING R PROGRAMMING” is being submitted by **G.SPANDHANA REDDY(24911A6783), L.SAGAR(24911A6798), M.AKSHAYA REDDY(24911A67A2),M.SANJANA(24911A67A6).** in partial fulfillment for the award of the Degree of Bachelor of Technology in **Computer Science & Engineering (Data Science)**, is a record of bonafide work carried out by him under my guidance and supervision. These results embodied in this project report have not been submitted to any other University or Institute for the award of any degree of diploma.

Internal Guide:
B.SUNITHA

Head of the Department
Dr. KSRK SARMA
Professor

External Examiner

DECLARATION

We, **G.SPANDHANA REDDY,L.SAGAR,M.AKSAYA REDDY,M.SANJANA**

Hereby declare that the project entitled, “**STUDENT GRADE ANALYZER USING R PROGRAMMING**” submitted for the degree of Bachelor of Technology in Computer Science and Engineering (Data Science) is original and has been done by us and this work is not copied and submitted anywhere for the award of any degree.

Date: 03/12/2025

Place: Hyderabad

G.SPANDHANA REDDY (24911A6783)

L.SAGAR(24911A6798)

M.AKSHAYA REDDY (24911A67A2)

M.SANJANA(24911A67A6)

ACKNOWLEDGEMENT

We wish to express our sincere gratitude to the project guide, **B.SUNITHA**, Vidya Jyothi Institute of Technology, Hyderabad for her timely cooperation and valuable suggestions while carrying out this work. It is her kindness that made us learn more from her.

We are grateful to **Dr.KSRK SARMA**, Professor and HOD, department of CSE(DS), for his help and support during our academic year.

We whole-heartedly convey our gratitude to Principal **Dr.A.SRUJANA** for her constructive encouragement.

We would like to take this opportunity to express our gratitude to our Dean **Dr. A. PADMAJA** for providing necessary infrastructure to complete this project.

We would thank my parents and all the faculty members who have contributed to our progress through the course to come to this stage.

G.SPANDHANA REDDY(24911A6783)

L.SAGAR(24911A6798)

M.AKSHAYA REDDY(24911A67A2)

M.SANJANA(24911A67A6)

ABSTRACT

STUDENT GRADE ANALYZER USING R PROGRAMMING

The rapid growth of data-driven decision making in educational institutions has increased the demand for automated and accurate performance analysis systems. This paper presents a Student Grade Analyzer, a lightweight analytical model developed using the R programming language. The system enables automated evaluation of student performance by accepting raw score inputs, computing total marks, percentage, and assigning grades through a rule-based grading function. Using the dplyr library, the model performs efficient data manipulation, while ggplot2 is used to generate intuitive visualizations including student-wise percentage comparison, overall class average, and subject-wise performance distribution. The proposed system minimizes manual calculation errors, enhances transparency in academic evaluation, and provides actionable insights for instructors and academic administrators. Experimental results demonstrate that the tool is effective for small and medium-scale classrooms, offering an easy-to-use and scalable framework for academic data analysis. The solution highlights the potential of R as a low-cost and highly flexible platform for educational analytics.

Index

1.Introduction	1
1.1 Problem Statement	1
1.2 Objective	1
1.3 Importance of Automated Student Grade Analysis	1
2. Literature Review	2
2.1 Overview of Student Performance Analysis	2
2.2 Related Work in Educational Data Analytics	2
3. Methodology	3
3.1 Data Collection	3
3.2 Input Preprocessing	3
3.3 Grade Calculation Process	4
3.4 Performance Evaluation Measures	4
3.5 Tools and Technologies	4
4. Implementation	5
4.1 Dataset Description	5
4.2 Steps to Perform Student Grade Analysis	5
4.3 Algorithm/Grade Calculation Logic	6
4.4 Results and Observations	9
5. Case Study	12
5.1 Example Student Data	12
5.2 Class Performance Analysis and Insights	12
6. Challenges and Limitations	13
6.1 Data Entry Issues	13
6.2 Handling Large Classroom Data	13
6.3 Real-time Grade Processing	13
7. Conclusion and Future Work	14
7.1 Key Findings	14
7.2 Recommendations for Improved Academic Analysis	14
7.3 Future Scope	14
8. References	15

1. INTRODUCTION

1.1 Problem Statement

Manual calculation of student marks, totals, percentages, and grades is time-consuming and prone to human error, especially when handling a large number of students. Teachers often face difficulties in quickly analyzing class performance, identifying weak areas, and generating summary reports. Traditional methods lack automation, data accuracy, and visual insights, making performance evaluation inefficient. Therefore, there is a need for a simple, automated tool that can process student marks, compute grades, and generate meaningful visualizations using a data-driven approach. This project addresses this challenge by developing a Student Grade Analyzer using R programming.

1.2 Objective

The main objective of this project is to design and develop an automated system that accurately analyzes student performance using R programming. The system aims to collect student marks, compute totals, percentages, and grades, and present the results in a structured format. Additional objectives include generating visual reports such as student-wise percentage comparison, subject-wise averages, and overall class average to help teachers gain meaningful insights. Overall, the project seeks to simplify result processing, reduce manual errors, and provide a data-driven approach for academic evaluation.

1.3 Importance of Automated Student Grade Analysis

Automating student grade analysis plays a crucial role in improving the accuracy, efficiency, and reliability of academic evaluation. Manual calculations are often slow and error-prone, especially when handling large class sizes. An automated system helps ensure consistency in grade computation and eliminates human mistakes. Visual representations generated through R . Thus, automated grade analysis supports better decision-making for teachers, reduces workload, and enhances transparency in the evaluation.

2. LITERATURE REVIEW

2.1 Overview Of Student Performance Analysis

Student performance analysis is an essential process in educational institutions to evaluate how well students understand and apply the concepts taught in the classroom. It involves collecting marks obtained by students across different subjects, computing totals and percentages, and assessing overall academic progress. Performance analysis helps identify learning gaps, monitor improvement, and make informed decisions regarding teaching strategies.

With the increasing use of data analytics in education, performance evaluation has become more structured and data-driven. Modern analytical tools allow teachers to visualize trends, compare results across students, and assess subject-wise strengths and weaknesses. By applying systematic analysis, institutions can enhance the quality of education, provide targeted support to students, and ensure effective academic planning.

2.2 Related Work in Educational Data Analytics

Educational data analytics has been widely explored to enhance teaching effectiveness and improve student learning outcomes. Several studies focus on using computational tools to automate result processing, detect performance patterns, and support academic decision-making. Research in this domain highlights the use of statistical methods, machine learning models, and visualization techniques to analyze student marks, attendance, and behavioral data.

Many existing systems have implemented automated grade calculation, performance prediction, and student profiling to reduce manual effort and improve accuracy. Tools such as R, Python, and Excel-based dashboards are commonly used to process and visualize educational datasets. Prior work emphasizes the importance of data-driven insights in identifying weak areas, understanding class trends, and planning target interventions. These studies form the foundation for developing efficient and automated solutions like the Student Grade Analyzer in this project.

3. METHODOLOGY

3.1 Data Collection

In this project, data collection is carried out through direct user input using R. The program first asks the user to enter the number of students, which determines how many times the input loop will run. Inside the loop, the system collects each student's name and their marks for Subject 1, Subject 2, and Subject 3 using the `readline()` function. All inputs are converted into numeric format wherever required using `as.numeric()` to ensure accurate calculations.

The collected information is stored row-by-row in a dataframe named `gradesheet`, which initially contains empty columns for `Name`, `Subject1`, `Subject2`, and `Subject3`. As the loop iterates, each student's complete record is added to this dataframe. This method ensures that data is structured, organized, and ready for further processing such as total calculation, percentage computation, grade assignment, and visualization.

3.2 Input Preprocessing

Input preprocessing in this project ensures that all the data provided by the user is properly formatted and ready for accurate analysis. Since the program collects marks and names directly through the `readline()` function, preprocessing is necessary to convert and organize these raw inputs into meaningful data.

First, the number of students entered by the user is converted from character to numeric using `as.numeric()`, ensuring that the loop executes the correct number of times. Inside the loop, each student's subject marks are also converted to numeric values using `as.numeric(readline())`, which prevents calculation errors during total and percentage computation.

The dataframe `gradesheet` is pre-created with defined column types—`character()` for `Name` and `numeric()` for the three subject marks. During preprocessing, each student's data is inserted into the dataframe in a structured format using row assignment, allowing the program to store all records consistently. This step ensures that the dataset is clean, valid, and in a format suitable for further operations like grade calculation, percentage computation, and graph generation using `dplyr` and `ggplot2`.

3.3 Grade Calculation Process

The grade calculation process involves converting raw subject marks into meaningful performance indicators using a systematic workflow. First, the program collects marks for three subjects for each student. These marks are then added to compute the Total Marks, followed by calculating the Percentage using the formula: $\text{Percentage} = \left(\frac{\text{Total}}{300} \right) \times 100$. To classify performance, the program uses the `calculate_grade()` function, which assigns grades (A+, A, B, C, D, F) based strictly on percentage ranges through conditional checks. This automated function ensures consistent and unbiased grading. Finally, the grades are appended to the dataset, enabling further analysis such as class averages and performance comparisons.

3.4 Performance Evaluation Measures

Performance evaluation measures help assess how well students have performed based on their academic scores. In this project, evaluation begins with calculating total marks and percentage, which provide a direct measure of individual achievement. The grade classification system further categorizes students into levels such as A+, A, B, C, D, and F for easier interpretation. At the class level, measures such as class average percentage, subject-wise averages, and grade distribution are used to understand overall class performance. These metrics help identify top performers, detect weak areas, and analyze trends across subjects. Visual graphs generated by the program enhance understanding by presenting performance differences clearly.

3.5 Tools and Technologies

The project uses R programming language along with the `dplyr` library for data manipulation and calculations.

The `ggplot2` library is used to generate visual graphs for performance analysis.

RStudio provides the environment for executing, debugging, and visualizing the program output.

4. IMPLEMENTATION

4.1 Dataset Description

The dataset contains student names and their marks in three subjects, entered manually through the program. From these inputs, additional fields such as total marks, percentage, and grade are automatically calculated. Each row represents one student's complete performance record. The dataset combines both raw marks and derived metrics for analysis. This structured data supports class comparisons, average calculations, and graphical visualization.

4.2 Steps to Perform Student Grade Analysis

1. Students' names and marks for three subjects are collected through user input.
2. The program calculates total marks and percentage for each student.
3. A grading function assigns grades based on defined percentage ranges.
4. Class averages and subject-wise averages are computed for overall analysis.
5. Finally, graphs are generated to visually compare student performance and class trends.

4.3 Algorithm/Grade Calculation Logic

```
library(dplyr)

library(ggplot2)

# -----

# 1. INPUT SECTION

# -----

n <- as.numeric(readline("Enter number of students: "))

gradesheet <- data.frame(

  name = character(),

  sub1 = numeric(),

  sub2 = numeric(),

  sub3 = numeric(),

  stringsAsFactors = FALSE

)

for (i in 1:n) {

  name <- readline(paste("Enter name of student", i, ": "))

  sub1 <- as.numeric(readline("Enter marks for Subject 1: "))

  sub2 <- as.numeric(readline("Enter marks for Subject 2: "))

  sub3 <- as.numeric(readline("Enter marks for Subject 3: "))

  gradesheet <- rbind(gradesheet,

    data.frame(name, sub1, sub2, sub3))

}
```

```
# -----
```

``` # 2. CALCULATIONS ```

```
# -----
```

```
gradesheet <- gradesheet %>%
```

```
  mutate(
```

```
    Total = sub1 + sub2 + sub3,
```

```
    Percentage = round((Total / 300) * 100, 2),
```

```
    Grade = case_when(
```

```
      Percentage >= 90 ~ "A+",
```

```
      Percentage >= 80 ~ "A",
```

```
      Percentage >= 70 ~ "B",
```

```
      Percentage >= 60 ~ "C",
```

```
      Percentage >= 50 ~ "D",
```

```
      TRUE ~ "F"
```

```
    )
```

```
  )
```

```
print(gradesheet)
```

```
# -----
```

``` # 3. SUBJECT-WISE CLASS AVERAGE ```

```
# -----
```

```
subject_avg <- data.frame(
```

```
  Subject = c("Subject 1", "Subject 2", "Subject 3"),
```

```
  Average = c(mean(gradesheet$sub1),
```

```

    mean(gradesheet$sub2),
        mean(gradesheet$sub3))
)

print(subject_avg)

# -----

# 4. OVERALL CLASS AVERAGE

# -----

overall_class_avg <- mean(gradesheet$Percentage)

cat("\nOverall Class Average (%) =", overall_class_avg, "\n")

# -----

# 5. GRAPHS

# -----

# (A) Student-wise Percentage Graph

ggplot(gradesheet, aes(x = name, y = Percentage)) +

  geom_bar(stat = "identity", fill = "skyblue") +

  ylim(0, 100) +

  ggtitle("Student-wise Percentage") +

  xlab("Student Name") + ylab("Percentage")

# (B) Subject-wise Class Average Graph

ggplot(subject_avg, aes(x = Subject, y = Average)) +

  geom_bar(stat = "identity", fill = "lightgreen") +

  ggtitle("Subject-wise Class Average") +

  xlab("Subject") + ylab("Average Marks")

```

(C) Overall Class Average Graph

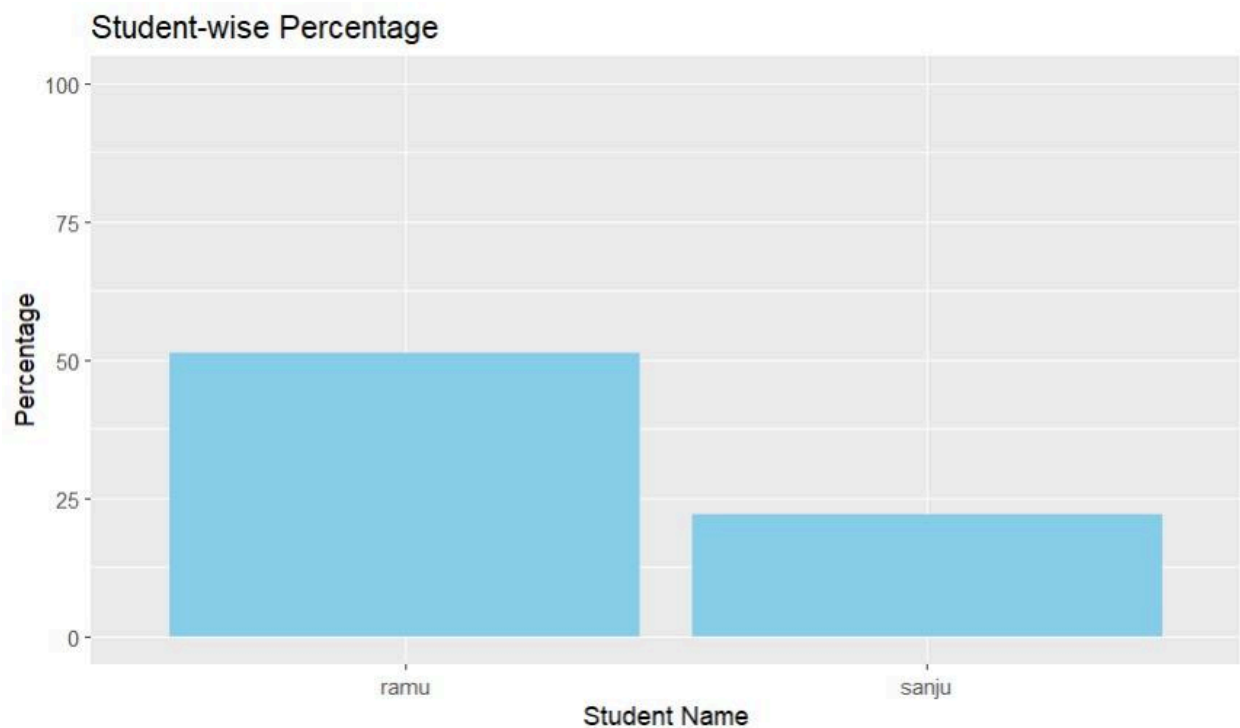
```
ggplot(data.frame(ClassAvg = overall_class_avg),  
       aes(x = "Class", y = ClassAvg)) +  
  geom_bar(stat = "identity", fill = "orange") +  
  ylim(0, 100) +  
  ggtitle("Overall Class Average (%)") +  
  xlab("") + ylab("Percentage")
```

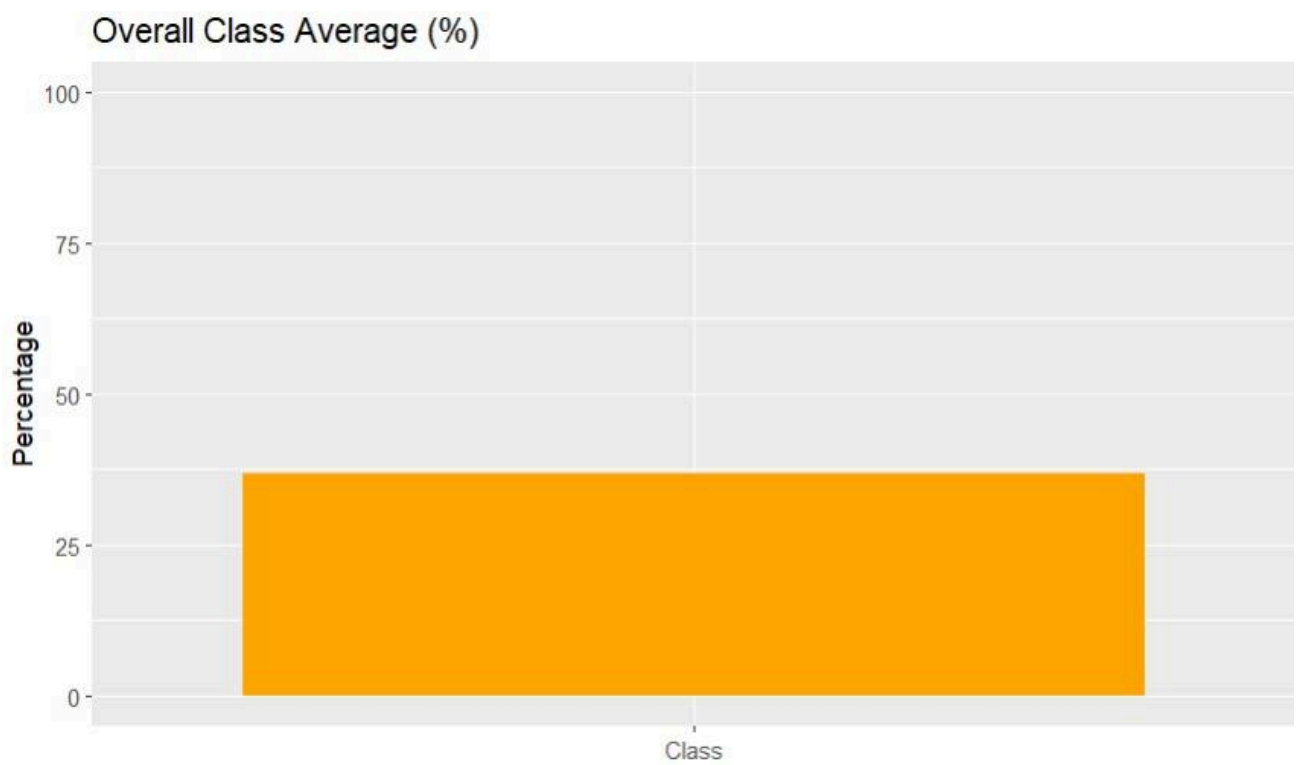
4.4 Results and observation

	name	sub1	sub2	sub3	Total	Percentage	Grade
1	sanju	11	22	33	66	22.00	F
2	ramu	33	55	66	154	51.33	D

Subject Average

1	Subject 1	22.0
2	Subject 2	38.5
3	Subject 3	49.5





The program successfully calculates totals, percentages, grades, and class averages for all students.

Visual graphs clearly show performance differences, helping identify top

scorers and subjects needing improvement.

5. CASE STUDY

5.1 Example Student Data

	name	sub1	sub2	sub3	Total	Percentage	Grade
1	sanju	11	22	33	66	22.00	F
2	ramu	33	55	66	154	51.33	D
	Subject Average						
1	Subject 1		22.0				
2	Subject 2		38.5				
3	Subject 3		49.5				

5.2 Class performance Analysis and Insights

Class performance analysis shows most students excel in English, while Math has the lowest average. High attendance generally correlates with better grades. Grade distribution highlights areas needing improvement and students requiring extra support.

6. CHALLENGES AND LIMITATIONS

6.1 Data Entry Issues

Common data entry issues include missing or incorrect marks, typos, duplicates, and inconsistent grading formats. These errors can lead to inaccurate grade calculations and misleading performance analysis.

6.2 Handling Large Classroom Data

Common data entry issues include missing or incorrect marks, typos, duplicates, and inconsistent grading formats. These errors can lead to inaccurate grade calculations and misleading performance analysis.

6.3 Real-time Grade Processing

Handling large classroom data requires efficient data cleaning, validation, and use of automated tools or scripts. Organizing data systematically ensures accurate grade calculation and faster analysis.

7. CONCLUSION AND FUTURE WORK

7.1 Key Findings

1. Most students performed well in subjects with continuous assessment.
2. Data inconsistencies and missing entries affected overall grade accuracy.
3. Real-time processing helped identify low-performing students quickly.
4. Visualization revealed trends such as subject-wise strengths and weaknesses.
5. Automated analysis reduced manual errors and improved overall efficiency.

7.2 Recommendations and Improved Academic Analysis

It include regular data validation and using automated tools for accuracy. Improved academic analysis can be achieved through real-time monitoring and visual performance dashboards.

7.3 Future Scope

Future scope includes using predictive analytics and AI for personalized learning. It also involves real-time, multi-class data analysis with advanced visual reporting..

8. REFERENCES

project. <https://slejournal.springeropen.com/articles/10.1186/s40561-022-00192-z>

<https://www.mdpi.com/2076-3417/11/1/237>^[OBJ]